W05 Requirement-to-Design Trace

ENGSE601 · Case X · SRS v1.1 + HLD v0.9 · evidence-based design review

Baseline: Case X SRS v1.1 (current official course README) and Case X HLD v0.9 draft under review. This is a design-document review, not a test run. Report support as Partially Supported/Not Verifiable where the HLD does not define enforceable behavior; no Pass/Fail is claimed.

1. Scope and source register

Artifact	Version / source	Use / limitation
Case X SRS	v1.1 · https://se-rmutl.github.io/engse601/materials/casex/ENGSE_CaseX_SRS_v1_1_Practical_TH.docx	Requirement baseline specified in current course repo; exact requirement wording and section references must be checked against full DOCX before peer review closes.
Case X HLD	v0.9 draft · https://se-rmutl.github.io/engse601/materials/casex/ENGSE601_CaseX_HLD_v0_9_Draft_for_Review_TH.docx	Design under review, not assumed correct. Read with the official visual guide.
HLD Visual Guide	https://se-rmutl.github.io/engse601/materials/casex/ENGSE601_CaseX_HLD_v0_9_Visual_Guide_TH.html	Used for described components C-01/C-02/C-05/C-08/C-10, interfaces and decisions D-02/D-03/D-04.
Worksheet trace anchors	ENGSE601_S1_Worksheet_TH.pdf, p.1: AC-X-05 and BR-X-02	Used as secondary cross-check of the requirement concepts; the bakery project portion of S1-B is separate and is not used as Case X evidence.

2. Design facts to carry into test design

Fact from HLD Visual Guide	What must be observable / verified
C-01 displays Readiness Summary and D-03 puts Overall Readiness calculation in the UI.	Observe readiness when required approvals are incomplete; confirm whether the same rule is enforced outside the UI.
C-02 hides Decision buttons for items not owned by the signed-in owner and sends decisions to C-05.	Try a non-owner request through the service/API path in a controlled test environment; hiding a UI button alone does not prove authorization.
C-05 records a Decision Record, writes to C-10 and calls C-08 for notification.	Trace decision -> persistence -> notification trigger; notification delivery channel/retry behavior must be specified.
D-02 gives availability cache a 15-minute interval; HLD section 7 lists multiple TBD values.	Freshness tolerance, interface/error format, performance threshold and retention policy are open questions unless the SRS/design owner confirms them.

3. Requirement-to-Design Trace

Trace / requirement basis	Design element and observation point	Support status	Fact / inference / missing evidence
TR-01 · FR-X-05: Overall Readiness topic (confirm exact clause in SRS v1.1)	C-01 Event Request UI, table 2; D-03. Observe Readiness Summary.	Partially Supported	Fact: Visual Guide links FR-X-05 to the readiness topic. Missing: exact SRS v1.1 wording/location and explicit service-side enforcement point.
TR-02 · BR-X-01: overall-status rule (confirm exact clause in SRS v1.1)	C-01 calculates overall readiness from loaded items; D-03 assigns calculation to UI. Observe status vs required-item state.	Partially Supported / risk open	Fact: HLD describes aggregate status at UI. Inference: other callers or stale loaded data may diverge without one authoritative rule. Missing: central enforcement contract and testable interface.
TR-03 · AC-X-05: must not be Ready while a required item is unapproved (worksheet p.1; verify full SRS v1.1)	C-01 Readiness Summary / D-03. Observe attempted Ready state while required item remains unapproved.	Partially Supported / Not fully verifiable	Fact: HLD places aggregate calculation at C-01. Missing: exact SRS v1.1 location, server-side rejection behavior and complete rule across entry paths. No runtime test claimed.
TR-04 · BR-X-02: owner may decide only assigned items (worksheet p.1; verify full SRS v1.1)	C-02 hides decision button; I-04 sends decision to C-05; D-04 relies on UI hiding. Observe direct non-owner decision attempt.	Partially Supported / authorization boundary unclear	Fact: HLD describes UI hiding. Inference: direct service call may bypass UI unless C-05 checks ownership. Missing: explicit C-05 permission rule and rejection contract.

4. Findings (Fact / Inference / Missing Evidence)

Finding	Fact from design	Inference / impact	Missing evidence
HF-01 / D-03	Overall Readiness is calculated in C-01 from loaded items.	If rule is only enforced in UI, other call paths may differ; acceptance test oracle not established for each path.	Exact SRS v1.1 clause; service contract; central rule/owner decision.
HF-02 / D-04	C-02 hides unavailable Decision buttons; C-02 sends decision to C-05.	UI hiding alone is not proof of authorization; possible direct-call design risk, not confirmed runtime defect.	Explicit C-05 ownership check, rejection response and permission test.
HF-03 / D-02	Availability cache is set to 15 minutes.	Staleness may affect readiness; acceptable age is not known here.	SRS freshness target and cache invalidation policy.
HF-04 / HLD §7	Visual Guide lists interface/error format, performance target TBD-X-01, notification retry/channel TBD-X-07 and retention TBD-X-08.	Without these, test oracle and NFR tests may not be fully defined.	Requirement/design owner decisions and exact source clauses.

5. Design Risk Register

ID / level	Risk / root cause	Impact on testing	Proposed action	Owner/status
DR-01 / High	D-03 places readiness calculation in UI.	Cannot show AC-X-05 holds across all call paths from UI behavior alone.	Decide authoritative rule and observation point; document service response.	Requirement/design owner: not assigned; Open
DR-02 / High	D-04 uses UI button hiding as permission control.	Direct-service authorization cannot be concluded from UI behavior.	Confirm C-05 checks ownership for every decision and returns defined rejection.	Security/design owner: not assigned; Open
DR-03 / Medium	D-02 cache interval is 15 minutes without confirmed freshness target.	Stale data boundary is not testable until a tolerance is stated.	Confirm freshness/invalidation threshold and corresponding test.	Requirement owner: not assigned; Open

6. Open questions / RTM handoff

ID	Open question	Evidence required / status
DQ-01	Does every trace row reflect the exact section/wording in the full Case X SRS v1.1?	Open: peer reviewer must compare each FR-X-05/BR-X-01/AC-X-05/BR-X-02 reference to the DOCX, record exact clause and correction if needed.
DQ-02	Should authoritative Overall Readiness be computed/enforced in an application service in addition to the UI?	Open: requirement/design owner decision and service rule/observable response needed.
DQ-03	What must C-05 validate when an owner submits a decision?	Open: authorization/ownership policy, rejection response and controlled permission test required.
DQ-04	What freshness threshold and invalidation rule apply to availability cache?	Open: source requirement and owner decision needed.

7. Handoff / peer review

Current support conclusion: Partially Supported / Not Verifiable for several trace rows. This is a finding about the HLD v0.9 draft, not a sign-off or runtime defect. Before closing W05, a teammate must verify exact clauses against SRS v1.1 and HLD v0.9, add reviewer/date, record findings/disposition/re-check, and update AI-LOG.md. No Pass/Fail test result is claimed.

8. Transfer note to ENGSE212 — THE ONE Bakery

Transfer only the review method, risk-based trace structure and open-question discipline. Do not copy Case X components or requirement IDs into the bakery baseline. The bakery project still needs its own approved Proposal/SRS and design source registered in INPUTS; keep these as Not Ready until the project owner provides them.

AI record: AI helped organize trace rows and review questions. The team must independently verify the source clauses, accept/change/reject proposals, and document the actual peer review.